Troubleshooting the Caterpillar 3116 Fuel System

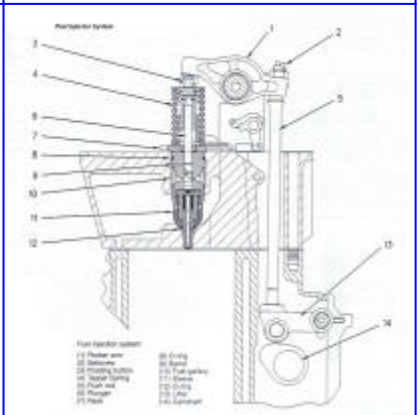
(1) Found the fuel solenoid on the fuel pump and make sure it clicked open when the key was turned on.

Put about 5psi of air pressure in the tank and check for fuel leaks, and found none.

(1) CAT 3116 Fuel System

(2) Fuel Filter Autopsy

(3) Fuel Samples; Left is from the bottom of the fuel tank, and right is out of the fuel filter.



(4) Caterpillar mechanical fuel injector detail.

(2) Disconnected the fuel out line from the tank a took a sample of the fuel, and also took a sample from the fuel filter and cut the filter apart to look for debris.

(2) This was some worrisome looking diesel. I am completely stumped as to what the white stuff is floating on the fuel tank. I the posted the photo the Yahoo origamiboats group. Turns out only the top layer is diesel with lots of water in it, and the bottom layer is mostly antifreeze.

So now I'm wondering what idiot put antifreeze in the fuel tank. I printed out the photo and took it to MHC, a local truck service station and and Ernie confirmed it was water. When I told him it was from a CAT 3116 he immediately asked if it had used more coolant lately.

(4) The 3116 as well as other CAT engines use mechanical injectors. They are located between the intake and exhaust valves and the work on the same principle. A push tube pushes a rocker arm and the other side of the rocker pushes on the top of the injector which compresses the fuel that is then injected into the cylinder. The fuel is delivered to each injector through a passage that is drilled though the center of the head. Because the work to compress the fuel is done at the injector the the fuel pressure from the pump and through the rail is only 29 to 58 psi depending on the engine speed. The coolant too of course travels through the head and actually passes around each injector that is sealed only by a thin brass sleeve which is pressed into the head. When the sleeve cracks or comes unsealed the fuel leaks into the coolant when the coolant pressure is low, and coolant leaks into the fuel when the coolant is hot and the pressure is high.

With some water in the fuel the engine was hard to start, but once the tank had an inch of antifreeze in the bottom, then nothing but antifreeze was being delivered to the engine and she died completely.

I cleaned out the fuel tank. Then filled a hose full of clean fuel and connected one end to the fuel line out from the filter and the blew air into the other in order to push clean fuel through the pump and rail. With the return line directed into a beer bottle I could get the engine to run just enough to move her out of the way until I get my gantry built when and can then pull the engine.

The gantry made easy work of lifting the cab, pulling the engine and replacing the head with a remanufactured head.

June 2008 - The gantry is up now and the engine is out. After a few evenings of disconnecting and labeling things for their inevitable reconnection we lifted the engine out of the truck. I then then rolled it around on the gantry until my neighbor Travis told me to stop showing off. I cobbled together a quick engine stand and set her down on a trailer so I can later drive her to a real mechanic in order to set the fuel injector control rack and valve settings.

After much discussion we decided to get a remanufactured head. The brass injector sleeves can be remove and new ones pressed back in, but that is another special CAT tool and the estimated hours brought the price to the same as the $800 remanufactured head. Add on 6 injectors, 3 destroyed injector push rods and rod adjuster ends, a new injector control rack as recommended, along with gaskets. The total damage was $2145 plus tax. The 3 destroyed injector push rods are likely due to the injectors locking up, because of the water. Luckily the rocker arms for the injectors appear to be in good shape. We first tried to get the injectors tested, but no one locally had the proper test equipment.

Setting the Fuel Injectors

With the new head bolted on the injector control rack and valve rockers arms are next and then it is time to adjust everything. If you have a fleet of CAT 3316's you will want the $1200 tool kit for setting the injectors. Or you can take it to a mechanic with the proper tools or you can take it to do what I did, and find an experienced diesel mechanic that says he can do it without the kit. That is the guy that might teach you something that could come in very handy.

Governor and the Fuel Control Ratio

The throttle cable connect to the governor and the governor connects to the fuel injectors. The governors job is to smooth out throttle increases so too much fuel is not dumped into the cylinder before the rpm's and turbo charger have caught up to the higher speed. In order to this trick the governor has a tube connected to the intake manifold and that moves a diaphragm attached to an arm the limits the travel of the injector control rail. As the intake pressure increases the diaphragm moves and the fuel slowly increases. The Fuel Control Ratio is an adjustment of how far behind the injectors are compared to where you have the throttle peddle. You can test the diaphragm by removing small meal tube that runs from intake manifold to the side of the governor. Apply 20 to 30 psi of air pressure to the fitting in the governor. If the engine is in working order you can snap the throttle back and forth while you watch the exhaust. With air applied you should get a big bloom of black smoke and without air applied not much smoke. If the engine is not running then testing the governor requires a special tool. If the governor is out of adjustment CAT has a special bench test set, of course.

Fuel Injector Synchronization

John Whelan's blog: http://SchoolBusMechanic.blogspot.com is a great resource for diesel work and John is good about answering questions. If you have the special tools from Caterpillar then John's site is exactly what you need. If you do not have the special tools it will show you what you will have to do without.